

RS6241

High Performance Single-Chip 60 GHz MIMO Radar Sensor

with Wireless Connectivity (BLE5.3/802.15.4/Thread/Matter)

Product Brief Rev. 1.0

Features

- **FMCW mmWave Radar**
 - 57-66 GHz Sweep Frequency Coverage
 - 2 Transmit Channels and 4 Receive Channels
 - Integrated RFPLL, DPLL, Transmitter, Receiver, ADC and Baseband
 - 12.0 dBm Typical Output Power per TX
 - 10.5dB Typical NF per RX
 - -95 dBc/Hz typical Phase Noise at 1MHz
 - 10/20MHz IF Bandwidth
 - Up to 800MHz/us Ramp Rate
 - HWA for 1D/2D FFT, Static Clutter Remove and CFAR Operations
- **2.4GHz Wireless Connectivity**
 - Protocols: Supports Bluetooth LE 5.3, Bluetooth Mesh, IEEE 802.15.4, Thread 1.3, Matter 1.1, and 2.4GHz proprietary
 - Bluetooth LE PHY : 1Mbps, 2Mbps, Long Range S2 (500Kbps), S8 (125Kbps)
 - TX Max Output Power: 10dBm@BLE mode
 - RX Sensitivity: -95dBm @BLE 1Mbps
- **Application System**
 - Two 32-bit RISC-V CPU Cores with DSP, FPU (up to 256MHz) and I/D cache for Radar Post-processing, Wireless and Application
 - Security Engine: Secure Boot, Efuse Key Protection, SRAM and XIP Decryption on the fly, TRNG, AES128/SHA/RSA/ECC
- **Memory**
 - On Chip 551KB SRAM
 - External PSRAM Support (up to 16MB)
- **Host and Device Interface**
 - x3 UART
 - x2 QSPI with XIP
 - x2 I²C
 - x8 PWM
 - x10 GPADC
 - x1 CAN-FD
- **Clock Source**
 - 32/40MHz Crystal Oscillator for Clock Reference
 - Internal LSI and MSI RC Oscillator
 - Optional 32kHz Crystal for Low Power Operation
- **Power Management**
 - Single Power Supply from 1.7V to 5.5V
 - Built-in LDO Network for Enhanced PSRR
 - BOM-Optimized and Power-Optimized Modes
- **Low Power Property**
 - Power Modes: Shutdown, Standby, Idle and Active (Sensing, Processing and Connecting)
 - Fast Wakeup from Standby to Active
- **Package**
 - FCCSP: 6.7 x 6.7 mm² with 94pins
- **Temperature Range**
 - Operating Junction Temperature: -40°C to 105°C

Applications

- Motion/Micro-Motion/Presence detection
- Vital signs/health monitoring
- Occupancy detection
- People tracking/counting
- Posture and fall detection
- Industrial detection
- Drone/robot obstacle avoidance
- In-cabin sensing for left behind (LBC) systems

Description

The RS6241 is a single chip 60GHz 2T4R FMCW MIMO radar sensor with 2.4GHz wireless connectivity (BLE5.3/802.15.4/Thread/Matter) inside. The device is portioned into four subsystems, shown in Figure below:

- **mmWave Radar Subsystem:** This block includes all high-performance radar RF/analog and baseband. It also includes the chirp generator and tx-to-rx finite state machine. HWA is included in baseband, used to offloading specified radar processing such as 1D/2D FFT and CFAR.
- **2.4GHz Wireless Connectivity Subsystem:** This block contains whole 2.4G multi-protocol wireless RF, Mod/Demod, Baseband Protocol, and the data interacts with application CPU by internal data bus.
- **Application Subsystem:** It mainly includes two 32bit RISC-V CPU cores with DSP, FPU and I/D cache. CPU connects SRAM, peripheral interface, and other subsystems by AHB and APB bus. RISC-V CPU is used to run radar post-processing algorithm, wireless protocol and other system tasks.
- **Power and Clock Subsystem:** This chip is complex single chip SoC with multiple function systems, we use single power and clock subsystem to do centralized management. It includes several clock sources (including LSI, MSI, DCXO and SPLL), power blocks(LDOs, BG and Clamp) and temperature sensor.

RS6241 is specifically designed for high performance and low power radar sense system applications in industry. It contains four main power modes for low power management: shutdown, standby, idle and active. Its configuration and data acquisition are enabled with UART/SPI/I2C/CAN-FD digital interface with other devices.

Block Diagram

